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Association of calving factors and labor duration with postpartum uterine health in Holstein cattle

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ABSTRACT

The objective of this observational study was to investigate the association between the duration of labor, calving assistance, and uterine diseases, as well as estimate a reference time to be used as a guideline for providing calving assistance based on the health outcome of metritis. Holstein cows ($n = 496$) were enrolled approximately 3 wk before their expected calving date. Cameras were used to record calving behavior of a subset of cows ($n = 286$), where the occurrence of calving assistance (forced extraction lasting a minimum of 5 s), calving location, and pen movement during labor were recorded. Calf weight was recorded at birth, retained fetal membranes (RFM) was diagnosed 24 h after calving and metritis was diagnosed based on vaginal discharge measured at 6 and 12 DIM. Subclinical and clinical endometritis (SCE) was based on cytological examination and vaginal discharge at 40 ± 5 DIM. Duration of stage II labor was estimated as the time from the appearance of the amniotic sac until the calf was expelled. Calving was assisted for 14% of cows. Duration of labor ranged from 11 to 173 min (mean $\pm$ SD = 57.2 ± 32.0) in unassisted cows and 49 to 232 min (122.7 ± 51.2) in assisted cows. As calf weight increased, the duration of labor increased (slope, $m = 1.16 \pm 0.34$). For every 5-kg increase in calf weight, there was 1.47 greater odds of RFM (95% CI = 1.38–2.08). Cows assisted during labor were at 3.67 higher odds of metritis when compared with unassisted cows (95% CI = 1.92–7.05). A quadratic relationship existed between metritis and duration of labor for assisted cows, where the predicted probability of metritis was greatest at the shortest (49 min; 69.2%) and longest (232 min; 100%) durations of labor, but the lowest predicted probability of metritis (22.1%) was at approximately 130 min. Predicted probability of metritis in unassisted cows was not affected

by duration of labor. Neither duration of labor nor the occurrence of calving assistance was associated with the development of RFM or SCE. Providing assistance either too early or too late during stage II labor may result in an increased risk of metritis; the results of this study suggest that intervention 130 min after the appearance of the amniotic sac may be associated with a reduced incidence of metritis. Further research is required to determine how progression of labor and timing of intervention affects the development of uterine diseases.

Key words: calving, dystocia, uterine diseases

INTRODUCTION

The transition period, defined as the 3 wk before to the 3 wk after parturition, is critically important for dairy cow health, production, and profitability because most health problems occur during this time (Drackley, 1999). Because calving of dairy cows is an essential feature of the dairy industry for milk production and herd renewal, cows go through the transition period often, approximately every 390 to 407 d (Hare et al., 2006). Dystocia, or difficulty in calving, has been shown to have a negative effect on the welfare of cows as well as negative economic impacts on farms (Buckley et al., 2003; Mee, 2008). Dystocia can result in reduced production and fertility and increased cow morbidity, mortality, involuntary culling, and veterinary costs (Dematawewa and Berger, 1997; Mee, 2008). Furthermore, cows diagnosed with dystocia are more likely to develop uterine diseases such as retained fetal membranes (RFM), metritis, and endometritis, in addition to other transition diseases (Sheldon et al., 2006). In addition to its effects on cow health, dystocia is also characterized by its effect on calf outcomes, particularly stillbirth, trauma, and reduced neonatal viability. These outcomes highlight the implications of dystocia for both dam and calf welfare. Calves born to dams that experienced dystocia have greater odds of stillbirth, and increased risk of respiratory and digestive disease, as well as increased overall mortality

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(Lombard et al., 2007). Similar to dystocia, uterine diseases can result in reduced milk production, reproductive performance, and life expectancy due to an increased risk of culling during the lactation (Bell and Roberts, 2007; Wittrock et al., 2011; Giuliadori et al., 2013; Stangaferro et al., 2016). In fact, metritis not only affects cows in their current lactation, but research also suggests long-lasting consequences (e.g., uterine disease, infertility) within their next lactation as well (Borchardt et al., 2024). Because of this, there has been an increase in research aiming to determine risk factors for dystocia and uterine diseases and how proactive management practices can prevent their occurrences or allow for early detection or mitigation of possible negative effects.

Parturition consists of 3 stages: stage I, the cervix and birth canal begin to dilate, and the calf moves into position; stage II, the cervix is fully dilated, abdominal contractions begin, the calf moves through the birth canal and expulsion of the calf occurs; and stage III, expulsion of the placenta (Wehrend et al., 2006). Dystocia has often been defined as a prolonged duration of labor or assisted extraction of the calf at birth as a result of prolonged stages I or II of parturition (Mee, 2004). However, over time, the term dystocia has often been used to describe only cows that required human assistance during calving. This is due to the lack of a consistent definition for prolonged labor and when to provide assistance. Prolonged labor has been defined by others as a calving duration taking longer than 70 min (Schuenemann et al., 2011; Kovács et al., 2016), 80 min (Mee, 2004), or 120 min (Gundelach et al., 2009) after the appearance of the amniotic sac. However, these time references are arbitrary and are not well supported by research explaining why they were chosen. Furthermore, research is lacking regarding the relationship between the duration of labor and the development of uterine diseases.

According to Noakes et al. (2001) dystocia occurs when one or more of the following main components of calving are inadequate: fetal size and presentation, expulsive forces, and birth canal adequacy. Reported factors that influence dystocia and prolonged labor include calf birth weight, maternal pelvic size, sire, parity, twins, season, milk fever, gestation length, weight at service and calving, and body condition at calving (as reviewed by Mee, 2004). Similarly, reported risk factors for metritis and endometritis include dystocia, twins, RFM, stillbirth, abortion, gestation length, and prolapsed uterus, and endometritis has additional risk factors such as parity, season, and body condition during the transition period (Markusfeld, 1987; Gröhn et al., 1990; Correa et al., 1993; Kaneene and Miller, 1995; LeBlanc et al., 2002; Kadivar et al., 2014; Vieira-Neto et al., 2017). In addition, metabolic disorders, such as hypocalcemia, hyperketonemia and displaced abomasum, can increase

the risk of both metritis and endometritis (Dubuc et al., 2010).

Further understanding of the calving process and how it affects uterine health can improve calving management practices to prevent the development of uterine disease. Therefore, the objectives of this observational study were to investigate the association between factors around the time of calving, duration of stage II labor, and the development of uterine diseases in dairy cattle. In addition, we aimed to estimate a reference time to be used as a guideline for providing calving assistance based on the health outcome of metritis. We hypothesized that cows with longer duration of unassisted labor will be more likely to be diagnosed with uterine diseases and that providing assistance too early or too late will increase the risk of uterine diseases.

MATERIALS AND METHODS

This project was conducted at the University of British Columbia's Dairy Education and Research Centre in Agassiz, BC, Canada from May 2016 to March 2019. All procedures were approved by the University of British Columbia's Animal Care Committee (protocol #A14-0019 and #A18-0043).

Animals and Housing

A total of 567 Holstein dairy cows, 161 primiparous and 406 multiparous, were enrolled in this observational study at 21 ± 3 d before the cows' expected calving date. Throughout this manuscript, cows which were calving into their first lactation were considered primiparous, and cows calving to their second or greater lactation were considered multiparous.

Lactating cows were dried off 60 ± 5 d (mean $\pm$ SD) from their expected calving date and moved to a dry pen immediately after the last milking. Cows were then moved into a prepartum pen (12.5 m $\times$ 9.7 m) containing 12 deep-bedded sand freestalls approximately 21 d before their expected calving date. Stocking density of the prepartum pen never surpassed 100% of the available stalls. Cows were moved into an adjacent maternity pen when they showed physical signs of imminent calving (i.e., relaxation of sacrospinal ligament, tail raising, and enlarged udder; Miedema et al., 2011; Proudfoot et al., 2013). The maternity pen was bedded with straw, approximately 30 cm deep and with dimensions of 30 m $\times$ 3.15 m, was cleaned daily, and straw was added as needed. The straw was fully replaced every 1 to 2 wk. Within 24 h after calving, cows were moved to lactating cow pens with deep-bedded sand freestalls. River sand was replenished once every 2 wk. Rubber matting lined

the alleys and was scraped approximately 6 times per day using an alley scraper.

Cows were fed a TMR formulated for transition cows before calving and a lactating TMR formulated for high-production cows after calving to meet or exceed the requirements of Holstein cows weighing 620 kg, producing 45 kg/day of 3.5% FCM (NRC, 2001). Before calving, fresh feed was delivered once daily and pushed up an additional 4 times daily. After calving, fresh feed was offered twice daily and pushed up twice between feedings. Water and feed were available *ad libitum*. Postpartum cows were milked twice daily, once between 0500 and 0800 h and again between 1500 and 1800 h, in a conventional double-12 parallel milking parlor.

Upon enrollment, body condition and gait were scored. Body condition score was given on a 5-point scale from thin (1) to obese (5) by increments of 0.25 as outlined by Edmonson et al. (1989). Cows were then classified as below average (BCS < 3.25), average (BCS = 3.25 or 3.5), or above average (BCS > 3.5) based on 33rd percentiles. Gait scores were given at the same time on a 4-point scale from sound (1) to severely lame (4) in increments of 1 as outlined by Flower and Weary (2006), while cows were walking freely. The length of the dry period, gestation length, and calf birth weight were recorded for each cow; all calves were purebred Holstein. Calf birth weights were obtained using a digital scale within 6 h of birth.

Data Recorded Around Calving

Video was used to record the behavior of a subset of the cows ($n = 381$; 111 heifers and 270 multiparous cows) within the prepartum and maternity pens. The final subset of calving events used for video analysis were equally distributed throughout the months of the year. Stage II labor, the occurrence of calving assistance, presence of other cows in the maternity pen at time of calving, calving location (maternity vs. prepartum pen), and if cows were moved during labor (after the appearance of the amniotic sac) were recorded continuously via 5 cameras, one placed above the prepartum pen and 4 above the maternity pen (CCTV camera, model WV-BP330 with F1.4/2.5–186 mm lenses; Panasonic, Osaka, Japan). Red lamps were positioned close to the cameras so that cows were still visible under low lighting. The prepartum camera was used to capture cows' calving behaviors before being moved into the maternity pen. All calving behaviors were watched retrospectively.

The duration of labor was assessed using video recordings and was measured, in seconds, from the first appearance of the amniotic sac until the calf was expelled, for both assisted and unassisted cows. Cows in the prepartum pen were checked for physical signs of imminent calving, as described previously, 4 times per day at 0600,

1200, 1800 and 2200 h. Delivery of the calf was assisted by farm personnel based on the discretion of farm management, in addition to the decision to move the cows into the maternity pen. General farm practice was to pre-emptively move cows to the maternity pen before the appearance of the amniotic sac and to give assistance only after a period of 45 min lapsed during which no further progression of labor occurred, yet active labor had previously been viewed (i.e., strong abdominal contractions, amniotic sac). However, these practices were not strictly enforced and thus allowed for a variation in calving duration within this study. No cows required veterinarian assistance for parturition. If assistance was required, all personnel used palpation gloves, obstetrical lubricating jelly (General Lube, First Priority, Elgin, IL), warm water with a diluted iodine solution (Prepodyne Scrub, Kane Veterinary Supplies, Edmonton, AB) for washing, and, when necessary, obstetrical chains. Assisted calving was defined as forced extraction of a calf via manual pulling or use of calf chains from one or more farm personnel lasting a minimum of 5 s. Calf birth weight and stillbirths (born dead or died within 24 h after birth) were recorded. Calving location (maternity pen vs. prepartum pen), presence of other cows in the maternity pen at the time of calving, and if the cow was moved during stage II labor (after the appearance of the amniotic sac) were recorded from video recordings. Three observers were used for video observation. To determine the reliability of video observations between observers, intraclass correlations were determined for 5 cows. The intraclass correlation coefficient ranged from 0.92 to 0.99, which is considered to be excellent (Hallgren, 2012).

Disease Diagnosis

Cows were visually inspected for fetal membranes appearing from the vulva approximately 24 h after calving, if fetal membranes were still present cows were diagnosed as having RFM. Supportive care for RFM was at the discretion of farm management. Vaginal discharge (VD) and temperature were evaluated for each cow at 6 and 12 d postpartum for the diagnosis of metritis. Examinations took place immediately following the morning milking and rectal temperature was recorded before VD assessment. Before vaginal examination, the vulva was thoroughly cleaned with a soft-bristled brush soaked in an iodine solution (Prepodyne Scrub, Kane Veterinary Supplies, Edmonton, AB) diluted in warm water to remove feces that could introduce bacteria into the vaginal canal. Discharge was evaluated using a Metrichick device (Simcrotech, Hamilton, New Zealand), which was inserted into the vaginal canal up to the cervix. The device was cleaned with the iodine solution between uses. The appearance and smell of the VD was evaluated using

the following scale adapted from Williams et al. (2005): 1 = no discharge or clear mucus; 2 = cloudy mucus with <50% purulent; 3 = purulent >50%; and 4 = putrid (red/brown color, watery, foul smelling). Cows were classified as having metritis if they had at least one VD score of 4 with and without a fever ($\geq 39.5^{\circ}\text{C}$) either at 6 or 12 d postpartum. The VD of cows that still had a fetal membranes attached at 6 d postpartum was not evaluated.

At 40 ± 5 d postpartum, following vaginal examination for VD as previously described, a cytological examination was performed via an endometrial cytobrush (CooperSurgical Inc., Trumbull, CT). In brief, the vulva of each cow was cleaned of any debris and the cytobrush was inserted into the vagina and then through the cervix. The cytobrush was threaded onto a steal rod and covered with an outer steel tube, which was all covered with a plastic chemise sanitaire. Once the rod was securely within the opening of the cervix, the chemise sanitaire was punctured, and once the rod was then within the uterus the cytobrush was exposed. With contact of the uterine wall, the cytobrush was extended from inside the steel tube and rotated one full turn to collect a sample. The cytobrush was then retracted inside the steel tube before removal from the uterus so as to not contaminate the sample. Slides for cytological examination were prepared on the farm by rolling the cytobrush on a clean glass microscope slide and staining using a Romanowsky stain (Diff-Quick, Fisher Diagnostics, Middletown, VA); coverslips were applied once the slide was dry using slide glue (Fisher Chemical Permout Mounting Medium, Fisher Scientific, Fair Lawn, NJ). Each slide was examined at $400\times$ magnification using a light microscope (Nikon Eclipse E200, Nikon, Tokyo, Japan) to perform a differential cell count of 300 (PMN and endometrial cells), done in 3 counts of 100 and then averaged, by one trained observer. Subclinical endometritis (SCE) was defined as the presence of $\geq 10\%$ PMN, without the presence of purulent discharge (>50%), based on Sheldon et al. (2006). To prevent misdiagnosis of a localized vaginal infection as clinical endometritis (CE), CE was diagnosed as purulent discharge (>50%) and presence of $\geq 10\%$ PMN, based on Sheldon et al. (2006). All additional health information, including diagnoses of milk fever, displaced abomasum, or culling, were collected by the dairy herd personnel with the assistance of the herd veterinarian and confirmed and recorded by the project leader using the on-farm Dairy Comp 305 software (Valley Agricultural Software, Tulare, CA).

Statistical Analysis

The required sample size of 61 cows with assistance at calving was estimated to detect a 24 percentage-point difference in metritis prevalence between cows with and

without calving assistance (77% vs. 53%; Viera-Neto et al., 2016), assuming 95% confidence, 80% power, and a 1:3 ratio between groups, while accounting for potential confounders (Dohoo et al., 2003; Leon and Heo, 2009).

All statistical analyses were performed in SAS Studio (version 3.8; SAS Institute Inc., Cary, NC). Before all analyses, data were checked for normality using the UNIVARIATE procedure and probability distribution plots; extreme outliers in unassisted calving behavior (greater than 3 interquartile ranges beyond the third quartile of the dataset) were considered to be a result of error and were removed ($n = 4$). To focus on risk factors of uterine disease in dairy cattle, 67 cows were excluded from the final analysis due to postpartum illnesses, twinning, or culling, which included cows diagnosed with milk fever ($n = 24$), displaced abomasum ($n = 9$), severe lameness (gait ≥ 4 ; $n = 11$), twinning ($n = 18$), culled immediately after calving ($n = 2$), or multiple conditions ($n = 3$), resulting in a total of 496 cows remaining for the analysis of risk factors for diagnosis of RFM, metritis, SCE, and CE. Additionally, 15 cows were missing RFM diagnosis, 7 cows were missing metritis diagnosis, and 73 cows were missing SCE and CE diagnoses. Cows diagnosed with RFM were removed from the metritis and endometritis models ($n = 45$) because RFM diagnosis was an intervening variable that explained metritis diagnosis. Because of the lack of cows with CE ($n = 11$), SCE and CE were grouped together into the new category of subclinical and clinical endometritis (**ENDO**). Final models, including the number of observations within these models have been summarized in Table 1.

A subset of 381 cows were recorded using video observation for duration of labor, after the exclusion criteria detailed in the previous paragraph, 347 cows remained. A total of 61 cows did not have complete duration of labor data as a result of unforeseen technical difficulties associated with the cameras (i.e., power outages, poor lighting, other cows blocking the view of calving cows) and therefore were removed from analyses assessing the duration of labor, leaving 286 calving events (48 assisted and 238 unassisted) for analysis. Cows that were assisted during parturition were also excluded from models assessing the duration of labor due to assistance being a confounding effect. In total 238 unassisted cows were used to assess duration of labor, and 280, 256, 228 cows (both assisted and unassisted) were used when assessing duration of labor as a risk factor for RFM, metritis, and SCE, respectively. Of the 286 complete calving events, a total of 5 stillbirths remained. Therefore, there were not enough stillbirth data to accurately analyze the association between calving factors and stillbirths. Final models, including the number of observations within these models have been summarized in Table 1.

Table 1. Description of final logistic and linear mixed regression models, with number of observations used within each model; for all models, cow was the experimental unit

Outcome*	Number of observations used	Independent variable	Direction of association	P-value
Duration of labor (min) ^{1,2,3}	201	Calf weight Parity (primiparous vs. multiparous)	Positive Negative	<0.01 0.10
Duration of labor (min) ^{1,2,3}	148	Days dry	Positive	0.07
RFM ^{4,5}	352	Calf weight BCS before calving	Positive Positive	0.03 <0.001
Metritis ^{1,5,6}	351	Calving assistance	Positive	<0.001
Metritis ^{1,5,6}	227	Duration of labor	Positive Quadratic	<0.01
Subclinical and clinical endometritis ^{5,7}	248	Metritis Parity (primiparous vs. multiparous)	Positive Negative	0.05 0.05

¹Due to collinearity and data subsets (e.g., duration of labor) two final models were reached.

²Duration of labor was defined as the time (minutes) from the appearance of the amniotic sac until the calf was expelled and only included cows that were unassisted.

³Models assessed using linear mixed models.

⁴Retained fetal membranes was defined as the presence of fetal membranes appearing from the vulva at approximately 24 h after calving.

⁵Models assessed using logistic regression.

⁶Metritis was diagnosed by assessing VD; those with putrid discharge (red/brown color, watery, foul smelling) at 6 or 12 DIM.

⁷Subclinical and clinical endometritis was classified as PMN $\geq 10\%$ at 40 ± 5 d postpartum, with or without purulence ($>50\%$), respectively.

Risk factors for the outcome of RFM, metritis, and ENDO were analyzed using logistic regression with cow as the experimental unit. To assess RFM, models included the categorical independent variables calving assistance (assisted or unassisted), parity (primiparous or multiparous), BCS before calving (below average, average, above average), and calving location (prepartum or maternity pens), in addition to the continuous independent variables of duration of labor (minutes), dry period length (days), gestation length (days), and calf weight (kilograms). Risk factors for metritis and ENDO were analyzed using logistic regression with cow as the experimental unit and including the same independent variables as described for RFM, with the addition of the categorical variable of metritis diagnosis included as a possible risk factor for ENDO. For RFM, metritis, and ENDO, odds ratios were used to report significant associations. Probability of metritis in relation to the duration of labor in assisted and unassisted cows was calculated with a regression equation consisting of linear, quadratic, and cubic effects of the duration of labor. Frequency tables were used to determine the proportion of assisted and unassisted cows that were diagnosed with metritis.

To assess factors associated with the duration of labor, duration of labor was rounded to the nearest minute and multivariable linear regression models, using cow

as the experimental unit, were constructed including the categorical independent variables of parity, BCS before calving, calving location, presence of other cows in the maternity pen at the time of calving (yes or no), and moved during labor (yes or no), in addition to the continuous independent variables of dry period length, gestation length, and calf weight. The estimates of categorical variables from the models were reported as LSM $\pm$ SEM.

Throughout the analysis, the following groups of variables were assessed in separate models due to collinearity: calving assistance, calving location, and duration of labor; dry period length, gestation length, and calf weight; calving location, presence of other cows in the maternity pen, and moved during labor. For all multivariable models, manual backward stepwise elimination was used, where variables were retained with $P \leq 0.15$. Significance was set at $P \leq 0.05$ and tendency at $0.05 > P \geq 0.1$.

RESULTS

Duration of labor ranged from 11 min to 173 min (mean $\pm$ SD = 57.2 ± 32.0 min; $n = 238$) in unassisted cows and 49 min to 232 min (122.7 ± 51.2 min; $n = 48$) in assisted cows. Duration of calving assistance ranged

from 6 s to 688 s (mean $\pm$ SD: 124.3 $\pm$ 156.0 s). Calving assistance occurred in 14% of cows. Distribution of explanatory variables, along with means and SD for continuous variables, hypothesized to be associated with duration of labor and uterine diseases are presented in Table 2. Factors associated with the duration of labor of unassisted cows were calf weight, days dry, and parity. As calf weight increased, the duration of labor increased (Figure 1; slope (m) $\pm$ SEM; 1.16 $\pm$ 0.34; $P < 0.01$). Although gestation length was not associated with duration of labor, as the length of the dry period increased the duration of labor tended to increase (m = 0.64 $\pm$ 0.33; $P = 0.07$). Primiparous cows tended to have a shorter duration of labor when compared with multiparous cows (LSM $\pm$ SEM; 50.4 $\pm$ 3.7 vs. 57.6 $\pm$ 2.5 min; $P = 0.10$). Duration of labor was not associated with gestation length, BCS, location of calving, whether the cow was moved during labor, or whether other cows were present in the maternity pen at the time of calving.

The prevalence of RFM for the study was 9% (Table 2). Calf weight and BCS were associated with RFM diagnosis. For every 5-kg increase in calf weight, there was 1.47 greater odds of RFM (95% CI = 1.38–2.08; $P = 0.03$). Cows with above average BCS had 3.7 higher odds of RFM when compared with average BCS cows (95% CI = 1.56–8.94; $P < 0.001$), and cows with below average BCS had 0.43 lower odds of RFM when compared with average BCS cows (95% CI = 0.15–0.128; $P < 0.01$). Duration of labor, calving assistance, parity, calving location, gestation length, and dry period length were not associated with RFM.

The prevalence of metritis for the study was 23.7% (Table 2); 83.7% of cows diagnosed with RFM were also diagnosed with metritis. Factors associated with metritis diagnosis, when excluding cows with RFM, were calving assistance and duration of labor. Cows that were assisted during calving had 3.67 higher odds of metritis when compared with unassisted cows (95% CI = 1.92–7.05; $P < 0.001$). Furthermore, the proportion of cows diagnosed with metritis was lower in unassisted cows when compared with assisted cows (Figure 2; $P < 0.001$). For every 10-min increase in the duration of labor, there was 1.13 higher odds of metritis (95% CI = 1.05–1.21; $P < 0.01$). Parity, calving location, dry period length, gestation length, calf weight, and BCS had no association with prevalence of metritis.

Duration of labor had a quadratic effect on the predicted probability of metritis in assisted animals, where the predicted probability of metritis in assisted cows was greatest at the shortest (49 min; 69.2%) and longest (232 min; 100%) durations of labor, but the lowest probability was found between 133 and 136 min (22.1%; $P = 0.03$; Figure 3). Although the probability of metritis in un-

assisted cows was not affected by duration of labor, a positive linear trend as observed, as shown in Figure 3.

The prevalence of ENDO was 18% for the study (Table 2), where cows diagnosed with RFM (30.6%) or metritis (35.5%) had a greater prevalence of ENDO. Factors associated with ENDO diagnosis were metritis and parity. Cows diagnosed with metritis at 6 or 12 d, or both, postpartum were at 1.92 higher odds of ENDO (95% CI = 1.00–3.75; $P = 0.05$) compared with nonmetritic cows. When compared with primiparous cows, multiparous cows were at 1.93 higher odds of ENDO (95% CI = 1.00–3.74; $P = 0.05$). Duration of labor, calving assistance, calving location, dry period length, gestation length, calf weight, and prepartum BCS were not associated with ENDO.

DISCUSSION

Although most cows experience normal calving, close monitoring before and during calving is necessary to identify calving difficulties in order to provide appropriate intervention. Inappropriate timing of interventions can have detrimental effects on both the dam and calf. Little research has investigated the relationship between the duration of labor and the development of uterine diseases; therefore, the objective of this study was to investigate the association between factors around the time of calving, duration of stage II labor, and the development of uterine diseases in dairy cattle. Although a few studies have researched the effect of improper timing of intervention on stillbirths and injury to the dam (Gundelach et al., 2009; Kovács et al., 2016), the cut-off times chosen for appropriate timing for intervention lack explanation. Results from the current study provide evidence that calving assistance occurring at approximately 130 min after the appearance of the amniotic sac is associated with a reduced probability of metritis postpartum. Assisting too early during stage II labor could result in an increased risk of metritis. Late intervention may also put cows at additional risk; however, it should be noted that our model has less power to detect this effect because an insufficient number of cows with extreme lengths of labor were included in the study. This time reference is similar to what Gundelach et al. (2009) reported: increased risk of stillbirths when assistance was provided after 120 min of stage II labor. Furthermore, Kovács et al. (2016) reported increased vaginal lacerations and RFM when assistance was provided too early within stage II labor (before 70 min). Previously reported management practices used to prevent metritis include adequate feed intake during the close-up period and appropriate conditioning of dry cows (Urton et al., 2005; Hammon et al., 2006; Huzzey et al., 2007; Neave et al., 2018); however, these management practices are long-term goals with limited use for inter-

Table 2. Distribution of the prevalence of RFM, metritis, and SCE, and risk factors hypothesized to be associated with duration of labor and uterine diseases in 496 cows

Item	Total cows measured	Classification	No. cows (%)	Mean	SD
Explanatory variable					
Parity ¹	496	Primiparous	160 (33)		
		Multiparous	336 (67)		
Calving assistance ^{2,3}	496	Assisted	68 (14)		
		Unassisted	428 (86)		
Prepartum BCS ⁴	462	Below average	94 (20)		
		Average	182 (40)		
		Above average	186 (40)		
Calving location ³	347	Maternity Pen	279 (80)		
		Prepartum Pen	68 (20)		
Presence of other cows in the maternity pen at time of calving	279	Yes	146 (52)		
		No	133 (48)		
Moved during labor ³	345	Yes	78 (23)		
		No	267 (77)		
Dry period length (d)	326			55.8	8.17
Gestation length (d)	496			278.9	4.80
Calf weight (kg)	448			42	5.70
Stillbirths ⁵	496	Yes	13 (3)		
		No	483 (97)		
Uterine diseases					
RFM ⁶	481	Yes	43 (9)		
		No	438 (91)		
Metritis ⁷	489	Yes	141 (29)		
		No	348 (71)		
Subclinical and clinical endometritis ⁸	423	Yes	347 (82)		
		No	76 (18)		

¹Cows which were calving into their first lactation were considered primiparous, while cows calving to their second or greater lactation were considered multiparous.

²Assisted calving was defined as the forced extraction, which lasted longer than 5 s, of a calf via manual pulling or use of calf chains from one or more farm personnel.

³Assistance of calf delivery as well as the decision to move cows into the maternity pen were at the discretion of farm management.

⁴Scored upon enrollment at 21 d before calving. Cows were classified as below average (BCS < 3.25), average (BCS = 3.25 or 3.5) or above average (BCS > 3.5) based on 33rd percentiles.

⁵Stillbirth was defined as the calf being born dead or died within 24 h of birth.

⁶Cows with RFM 24 h after calving.

⁷Metritis was classified as at least one VD score of 3 with a fever ($\geq 39.5^{\circ}\text{C}$) or VD score of 4 at either 6 or 12 d postpartum.

⁸Subclinical and clinical endometritis was classified as PMN $\geq 10\%$ at 40 ± 5 d postpartum, with or without purulence ($>50\%$), respectively.

ventions during calving. The current study demonstrated that proper timing of intervention is key to providing immediate aid in preventing or reducing risk of metritis postpartum. It must be noted that required calving assistance within this study was only minimal because the longest duration of human assistance lasted only 11.5 min and no cows required veterinary assistance. Even as such, the occurrence of assistance, irrespective of timing, was associated with an increased risk of developing metritis early postpartum. Although this study did not find an association of human assistance with the development of RFM or endometritis, it is plausible that the severity or length of human assistance was not sufficient to see these detrimental effects. Future research is needed to

better understand the association of the length and force of human assistance with uterine health outcomes.

Although calving assistance was left up to the discretion of farm management, the mean duration of labor in unassisted cows (57.2 min) was comparable to other studies. Schuenemann et al. (2011) reported the mean duration of stage II labor in unassisted cows to be 45.2 ± 12.2 min (mean $\pm$ SD) and Kovács et al. (2016) reported the mean duration to be 77.4 ± 6.2 min (LSM $\pm$ SEM) for cows in individual maternity pens and 52.6 ± 5.5 min for cows housed in a group maternity pen. Furthermore, calving assistance (extraction of calf using human force lasting more than 5s) occurred in 14% of the cows in this study. Within the literature rates of dystocia range from 1.5% to 30% (Mee, 2008; Tsousioti et al., 2024),

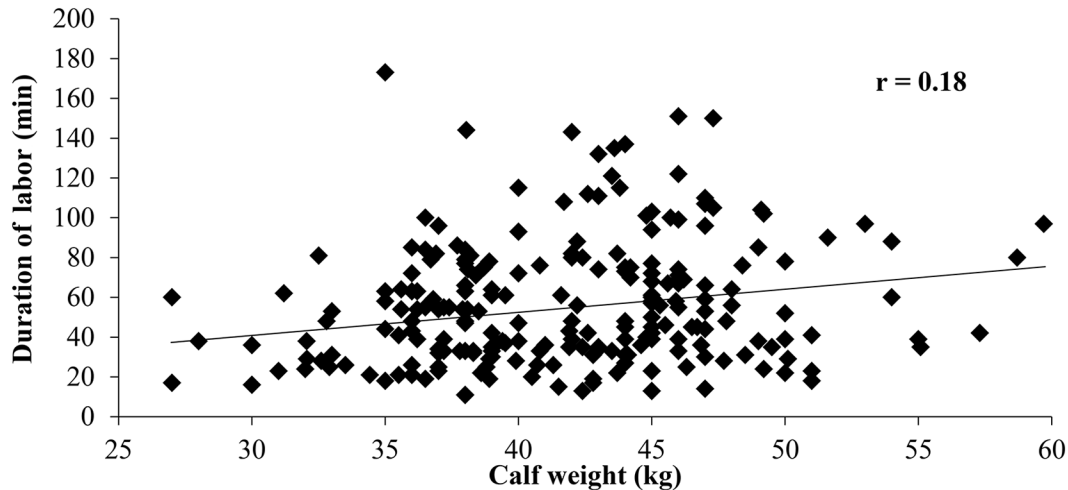


Figure 1. Scatter plot representing the correlation of calf weight with duration of stage II labor (appearance of amniotic sac until calf expulsion) for unassisted calving in Holstein dairy cows ($n = 224$; $P < 0.01$).

although these studies vary in dystocia definition (e.g., requiring surgical intervention, hard pull) and in breed studied. For a review of rates of dystocia over time, refer to Tsaousioti et al. (2024). Although duration of stage II labor and calving assistance did not have a direct effect on ENDO, it may be indirectly affected by timely intervention through the action of metritis, because ENDO was found to be increased in cows diagnosed with metritis. Moreover, other studies have reported increased odds of endometritis associated with both dystocia and metritis (Dubuc et al., 2010; Potter et al., 2010). These contrasting results are likely because of differences in the management protocols for providing assistance, the

definition used for dystocia, and different methods of diagnosing endometritis. For example, Dubuc et al. (2010) defined dystocia as requiring more than 2 people or a calf-jack, with the processes lasting longer than 15 min, but Potter et al. (2010) defined dystocia as requiring human assistance with no distinct definition. As previously mentioned, the definition of dystocia widely varies throughout the literature, for an in-depth review on these definitions refer to Mee (2008) and Tsaousioti et al. (2024).

As calf weight increased, duration of labor increased, and the odds of RFM also increased. It has been estimated that the rate of dystocia increases by 13% for ev-

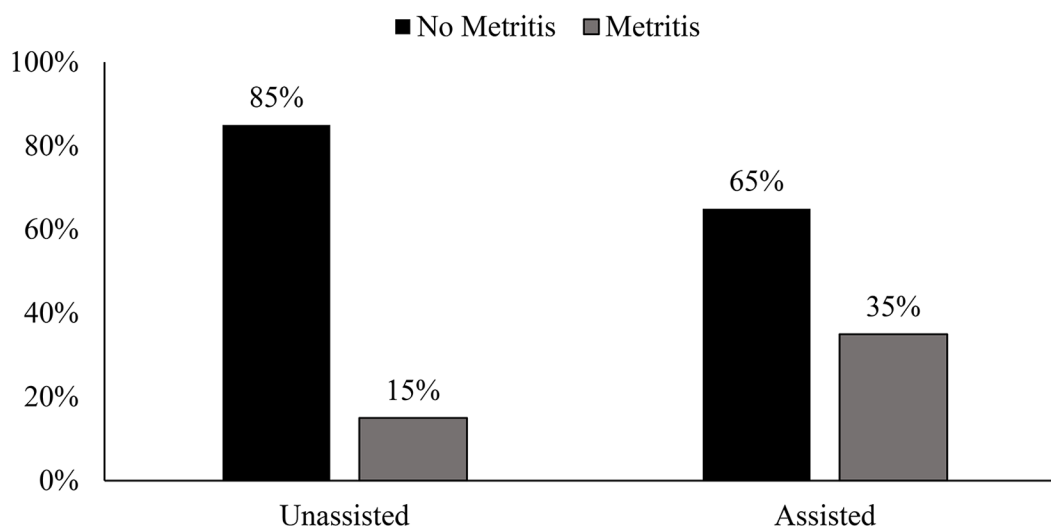


Figure 2. Proportion of cows diagnosed with metritis at 6 or 12 d postpartum ($P < 0.001$). A total of 78 cows were diagnosed with metritis (57 unassisted and 21 assisted at calving) and 354 cows were not diagnosed with metritis (315 unassisted and 39 assisted).

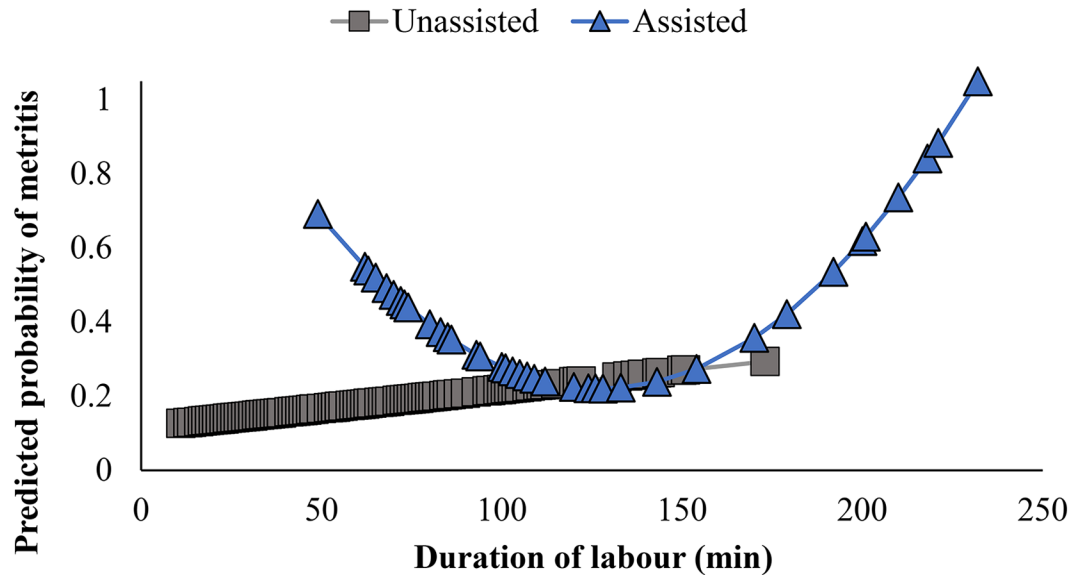


Figure 3. Effect of duration of stage II labor (appearance of amniotic sac until calf expulsion) in cows that were assisted ($n = 41$; $P = 0.03$) or unassisted ($n = 218$; $P = 0.20$) at calving on the predicted probability of metritis.

ery additional kilogram of birth weight above 40.3 kg, in Holstein cows (Johanson and Berger, 2003). Although a significant relationship existed between calf weight and duration of labor, calf weight was not a strong predictor for duration of labor in this study, as explained by a low r -value (Figure 1). Contrary to our results, Guard (1999) suggested that inappropriate timing resulting from delayed calving assistance can lead to RFM because of increased uterine inflammation. The association between calf weight and RFM in the current study could be the result of larger calves putting more pressure on the uterine wall and therefore causing inflammation, rather than from the calving duration or assistance itself.

Although gestation length was not associated with the duration of labor, dry period length tended to increase the duration of labor. This is likely due to longer days dry increasing the weight of the calf (Atashi et al., 2013); however, decreasing the number of days dry to reduce calving difficulties is not recommended because a dry period of 50 to 60 d, such as that used within this study, can optimize milk production and profitability postpartum (Schaeffer and Henderson, 1972). Neither gestation length nor dry period length were associated with metritis, ENDO, or RFM. Vieira-Neto et al. (2017) reported primiparous cows with short gestation lengths had higher rates of RFM and metritis, whereas multiparous cows with short or long gestation lengths had higher rates of dystocia, RFM, and metritis. Although it may appear that artificially inducing labor to occur at a normal gestation length of around 280 d could aid in reducing calving and postpartum complications, inducing labor can result in

unpredictable durations of labor and negative consequences on the calf and the dam (Barth, 2006).

Primiparous cows had a shorter duration of unassisted labor than multiparous cows. These results contrast with a previous study that reported primiparous cows to have a higher incidence of dystocia and increased odds of prolonged stage II labor when compared with multiparous cows (Gundelach et al., 2009). Schuenemann et al. (2011) reported no differences in parity for duration of unassisted labor. The different results found on the effect of parity on the duration of unassisted labor among studies (Gundelach et al., 2009; Schuenemann et al., 2011) is likely the result of the definitions of dystocia used. Furthermore, the lack of increased calving difficulties in primiparous cows in the current study may be associated with farm-specific management factors, including the age at which heifers are bred (12–14 mo), and the use of small sires often with sexed-semen to result in smaller, female calves.

It is generally accepted that overconditioning before calving predisposes cows to calving difficulties (Noakes et al., 2001), and underconditioning before calving predisposes cows to endometritis (Dubuc et al., 2010) and metritis (Giuliodori et al., 2013). Cows with below average BCS were at the lowest odds of RFM and metritis when compared with cows with average or above average BCS, whereas cows with above average BCS were at the highest odds of RFM. Although cows with below average BCS performed better in our study (less RFM and metritis), this could be due to a lack of truly thin animals in the herd (our lowest category of BCS included cows of BCS

<3.25 with only 1% of cows having a BCS ≤ 2.5). A BCS of 3 is considered ideal for dry cows prepartum (Dubuc et al., 2010) and has been suggested as the target BCS at calving to optimize health and production (LeBlanc, 2010).

The presence of other cows in the maternity pen and calving location was not associated with duration of labor in the present study. Kovács et al. (2016) reported a shorter duration of labor in unassisted calving and a lower prevalence of dystocia in cows housed in group maternity pens, such as those used in the current study, compared with individual maternity pens. Contrary to Kovács et al. (2016), Proudfoot et al. (2014) reported that dairy cows in individual maternity pens seek seclusion to calve and therefore calving in a maternity pen without a secluded area is not ideal. Although we found no association between being moved during stage II labor and the duration of labor, Proudfoot et al. (2013) found that moving cows during the late part of the stage I labor caused a delay in the second stage of labor. This difference in results may be the result of Proudfoot et al. (2013) moving cows before stage I labor rather than moving cows during stage II labor as in the current study. Furthermore, location of calving had no association with uterine diseases postpartum. Although hygiene of the calving environments and uterine bacteria flora were not measured in this current study, evidence is lacking indicating that hygiene of the calving environment influences the uterine bacteria flora and development of metritis or endometritis (Noakes et al., 1991; Potter et al., 2010); however, poor hygiene is generally accepted to increase the risk of uterine diseases in early postpartum. Although no association between calving in freestalls within the prepartum pen was found on the development of RFM, metritis, SCE, and CE, it is discouraged due to increased discomfort to the cow and calf, as well as safety risks for the calf and farm personnel.

Because this was an observational study conducted at a single research facility, the generalizability of the results to commercial herds with different management systems, cow genetics, or environmental conditions may be limited. The decision to provide calving assistance was made by farm personnel based on their judgment rather than using standardized intervention criteria, which may have introduced selection bias: Cows perceived to be at higher risk or progressing more slowly could have been assisted earlier, influencing both the duration of labor of assisted cows and subsequent disease outcomes. Additionally, the observational nature of the study impedes causal inference, and unmeasured variables such as hormonal status or individual cow temperament could have influenced both labor progression and disease susceptibility. Future research using controlled experimental designs across

multiple herds would help validate the reference time identified for intervention and further clarify the causal relationship between labor dynamics and uterine health.

CONCLUSIONS

Appropriate timing of calving intervention can aid in the prevention of metritis early postpartum. Specifically, providing assistance too early during stage II labor may result in an increased risk of metritis; therefore, we estimate 130 min, or roughly 2 h, after the appearance of the amniotic sac could be used as a reference point for providing calving assistance to reduce the risk of metritis. Although calf weight was associated with duration of labor and RFM, we were unable to find associations between uterine diseases and other important calving factors. Furthermore, duration of labor and calving assistance appeared to not be directly associated with ENDO, although appropriate timing of intervention may indirectly reduce the risk of ENDO by reducing the risk of metritis early postpartum. Future research should focus on how the timing of intervention with regards to progression of labor affects the dam, specifically with regards to uterine diseases. Furthermore, a collective definition of dystocia in which prolonged labor is defined is needed for more accurate comparisons between studies.

NOTES

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Nonstandard abbreviations used: CE = clinical endometritis; ENDO = cows with either SCE or CE; RFM = retained fetal membranes; SCE = subclinical endometritis; VD = vaginal discharge.

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